

الإختصاص والرمز : المعلوماتية: فرع البرمجيات (أ5306)
المادة : نظام هيكلية المستخدم / الموزع
المستندات المسموح بها: لا شيء

المعدل : (10)
المدة : ساعة ونصف

Treat the following questions:

1. Define the following terms: **(5 pts)**
 - a) TCP
 - b) UDP
 - c) URL
 - d) Host
 - e) RMI
2. Construct a detailed diagram that shows the flow of information between all the components of a 2-tier client/server architecture. **(4 pts)**
3. Define CORBA. Give four services provided by CORBA. **(3 pts)**
4. Explain the purpose of each instruction in the following Java program: **(8 pts)**

Line #	Java Code
1	import java.net.Socket;
2	public class SocketMethods {
3	public static void main(String[] args) throws Exception {
4	String rhn = "yahoo.com";
5	int rhp = 80;
6	System.out.println("\n Connecting to " +
7	rhn + " on port " + rhp + "...");
8	Socket socket = new Socket(rhn, rhp);
9	System.out.println("\n Connected to " + rhn +
10	" on port " + rhp);
11	System.out.println("\n getInetAddress(): " +
12	socket.getInetAddress());
13	System.out.println("\n getPort(): " +
14	socket.getPort());
15	System.out.println("\n getLocalAddress(): " +
16	socket.getLocalAddress());
17	System.out.println("\n getLocalPort(): " +
18	socket.getLocalPort());
19	socket.close();
20	}
	}

Traiter les questions suivantes :

1. Définir les termes suivants : (5 pts)
 - a) TCP
 - b) UDP
 - c) URL
 - d) Host
 - e) RMI
2. Construire un diagramme détaillé qui montre le flux d'information entre les différents composants d'une architecture client/serveur à deux niveaux. (4 pts)
3. Définir CORBA. Donner quatre services offerts par CORBA. (3 pts)
4. Expliquer l'utilité de chaque expression dans le programme Java suivant : (8 pts)

Line #	Java Code
1	import java.net.Socket;
2	public class SocketMethods {
3	public static void main(String[] args) throws Exception {
4	String rhn = "yahoo.com";
5	int rhp = 80;
6	System.out.println("\n Connecting to " +
7	rhn + " on port " + rhp + "...");
8	Socket socket = new Socket(rhn, rhp);
9	System.out.println("\n Connected to " + rhn +
10	" on port " + rhp);
11	System.out.println("\n getInetAddress(): " +
12	socket.getInetAddress());
13	System.out.println("\n getPort(): " +
14	socket.getPort());
15	System.out.println("\n getLocalAddress(): " +
16	socket.getLocalAddress());
17	System.out.println("\n getLocalPort(): " +
18	socket.getLocalPort());
19	
20	socket.close();
	}
	}